01: Intro Stats Review

Stat 230 - Spring 2026

Researchers examined the relationship between sleep duration and simple reaction time performance in healthy young adults in a controlled laboratory setting.

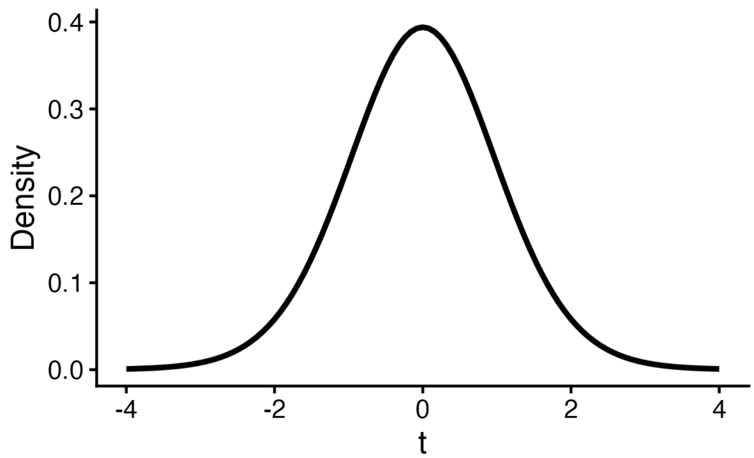
Sixty-five college-aged participants (ages 18-25) were recruited and randomly assigned to sleep either 4 hours per night (**low sleep**) or 8 hours per night (**high sleep**) over a one-week period. Each morning, participants completed a computerized simple reaction time task where they responded to visual stimuli as quickly as possible, with reaction times measured in milliseconds. The researchers are interested in whether there is a difference in reaction time for the low sleep vs the high sleep group.

	Sample size (n)	Sample mean ($\bar{x}$)	Sample SD (s)
Low Sleep	33	390 ms	45 ms
High Sleep	32	360 ms	40 ms

1. What is the null hypothesis for this test?

2. What is the alternative hypothesis for this test?

3. The researchers conduct a two-sample t-test and report a test statistic of $t = 2.84$. Below is a plot of the appropriate t-distribution for this test. Sketch how you would calculate the p-value for this situation.



4. How did you calculate this p-value in your last statistics course?

5. If the p-value you calculated was 0.02, what would you conclude about the null hypothesis?

6. Why did researchers randomly assign participants to a sleep condition?

7. Can you generalize these results to all young adults in the U.S.? Why or why not?